

Usage:

minimize [*options*](#)

Minimize is the command-line implementation of [Minimize Structure](#); it energy-minimizes molecule models, optionally holding some atoms fixed. [Amber](#) parameters are used for standard residues, and Amber's [Antechamber](#) module (included with Chimera) is used to assign parameters to nonstandard residues.

Before energy calculations can be performed, it is necessary to correct structural inconsistencies, add hydrogens, and associate atoms with force field parameters.

Minimize calls [Dock Prep](#) to perform [several tasks](#) to to prepare the structure(s).

[Dock Prep](#) may in turn call other tools ([AddH](#) and [Add Charge](#)). Currently, [Add Charge](#) (or the command [addcharge](#)) must be run prior to minimization because it plays a crucial role in assigning [parameters](#).

[Steepest descent](#) minimization is performed first to relieve highly unfavorable clashes, followed by [conjugate gradient](#) minimization, which is much slower but more effective at reaching an energy minimum after severe clashes have been relieved. Consult the [Minimize Structure](#) manual page for further details.

Options

Option keywords for **minimize** can be truncated to unique strings and their case does not matter. A vertical bar "|" designates mutually exclusive options, and default values are indicated with **bold**. Synonyms for true: True, 1. Synonyms for false: False, 0.

spec [*atom-spec*](#)

Minimize/move only the atoms in [*atom-spec*](#). All other atoms will be held in place automatically, without explicit use of [freeze](#). However, energy calculations will include the entire molecule models containing the specified atoms, unless [fragment true](#) is also used. All included models are treated as a single system. If [*atom-spec*](#) contains any spaces, it must be enclosed in single or double quote marks.

fragment true | false

Instead of the entire models, include in energy calculations only the

residues containing the [spec](#) atoms. Residues can only be included/excluded as a whole, and all included residues (even if from multiple models) are treated as a single system. Note, however, that any subset of the [spec](#) atoms can be [frozen](#) in place.

cache true | false

Whether to cache parameters such as partial charges for the current system (models or [fragment](#) to be included in energy calculations).

Caching allows reusing the data in subsequent rounds of minimization after simply updating the coordinates. Cached data for a system will be kept indefinitely unless Chimera detects a change in that system (atom addition or deletion); setting **cache false** helps to prevent overconsumption of memory.

prep true | false

Whether to call [Dock Prep](#) when parameters for the current system are not in the [cache](#). (If the parameters for the system are in the cache, [Dock Prep](#) will not be called, regardless of this setting.) The purpose of **prep false** is to avoid re-preparing structures that have already been prepared but not cached (for example, a different [fragment](#) than was previously minimized).

freeze none | selected | unselected | [atom-spec](#)

Which atoms (in addition to those [frozen automatically](#)) to hold in place during minimization: none (default), those which are [selected](#), those which are not selected, or those indicated with an [atom-spec](#).

Previously this option was required to freeze any atoms, but now that atoms other than the [spec](#) atoms are frozen automatically, this option is only needed in the case of:

1. restricting the calculation to certain residues with the [spec](#) option and [fragment true](#)
2. additionally freezing some of the [spec](#) atoms

This combination of options is useful for preventing long or distorted bonds at the junction between included and excluded residues. For example, minimizing a linker between two domains that are excluded from the calculation could significantly degrade the linker-domain bonds (since those bonds are ignored) unless the linker endpoint atoms are frozen in addition to the domains.

nsteps *N*

N (default **100**) is the number of steps of steepest descent minimization to perform before any conjugate gradient minimization.

stepsize *S*

S (default **0.02** Å) is the initial step length for steepest descent minimization.

cgsteps *M*

M (default **100**) is the number of steps of conjugate gradient minimization to perform after finishing any steepest descent minimization.

cgstepsize *D*

D (default **0.02** Å) is the initial step length for conjugate gradient minimization.

interval *I*

I (default **10**) is how often to update the display, in terms of minimization steps.

nogui true | false

Setting **nogui** to **true** suppresses the appearance of tool dialogs that might otherwise appear. [Dock Prep](#) and [AddH](#) are executed with their default settings (as shown when running **minimize** with **nogui false**), while [Add Charge](#) uses Amber **ff12SB** charges for standard residues ([details](#)) and for any nonstandard residues, the [method](#) last specified in its GUI (**AM1-BCC** if the GUI has not been used).

See also: [addh](#), [addcharge](#), [findclash](#), [Dock Prep](#), [Adjust Torsions](#), [Minimize Structure](#)

Usage:

addcharge std [spec [atom-spec](#)] [chargeModel *version*]

Usage:

addcharge nonstd [atom-spec](#) *net-charge* [method am1 | gas]

Usage:

addcharge [all [spec [atom-spec](#)] [chargeModel *version*] [method am1 | gas]]

Addcharge is the command-line implementation of [Add Charge](#). It assigns atomic partial charges and Amber/GAFF atom types as the [attributes](#) **charge** and **gaffType**, respectively. Atomic partial charges are included in saved [Mol2 files](#). The charges and Amber/GAFF atom types are used by other Chimera tools such as [Minimize Structure](#).

Charge assignment requires explicit hydrogens, which can be added beforehand with [AddH](#) (or the command [addh](#)). Structures often have problems such as truncated sidechains that can lead to non-integral net charges. The [Dock Prep](#) tool includes options for fixing such problems prior to hydrogen addition and charge assignment.

- Atoms in *standard residues* (water, standard amino acids, standard nucleic acids, and a few common variants and capping groups) are assigned charges and types from [Amber](#) ([details](#) for standard residues).

The keyword **std** (or **s**) indicates assignments should only be made to standard residues. If **spec** is used, the charge assignment will be limited to the entire model(s) containing the [specified](#) atoms.

The keyword **chargeModel** indicates Amber force field *version* ([details](#)):

- **ff12SB** or **12SB** (default)
- **ff02pol.r1** or **02pol.r1** (however, polarizability is not supported by [minimize](#), see [note](#))
- **ff03ua** or **03ua** - treats amino acid sidechain aliphatic hydrogens as if

collapsed onto the corresponding carbons; these “extra” hydrogens will be removed (however, the united-atom approach is not supported by [minimize](#), see [note](#))

- **ff03.r1** or **03.r1**
 - **ff99SB** or **99SB** (deprecated)
 - **ff99bsc0** or **99bsc0** (deprecated)
- Except for *monatomic ions*, which are simply assigned their net charges, atoms in *nonstandard residues* are assigned charges and [GAFF types](#) determined by [Antechamber](#), which is included with Chimera ([details](#) for nonstandard residues).

The keyword **nonstd** (or **n**) indicates assignments should only be made to nonstandard residues containing the [specified](#) atoms. All of these residues must be of the same type (with the same residue name and containing the same types of atoms connected in the same way). The *net-charge* for that type of residue must also be supplied; it can be zero, or a positive or negative integer. The charge calculation [method](#) can be **am1-bcc** (default, synonym **am1**) or **gasteiger** (synonym **gas**).

- The keyword **all** (or **a**, or the absence of arguments) indicates charges should be assigned to both standard and nonstandard residues. If atoms are [specified](#), the charge assignment will be limited to the entire model(s) containing the specified atoms. If nonstandard residues are found, a dialog for specifying their net charges will appear. The **chargeModel** for standard residues and charge calculation **method** for nonstandard residues can be specified as described above.

Explicit hydrogens are required for adding charges. [AddH](#) will be called as needed to add hydrogens.

If a nucleic acid chain has a 5' terminal phosphate, the user will be asked whether this group should be deleted; otherwise, its atoms will be assigned charges of zero (the Amber charge set lacks parameters for 5' phosphates).

Unrecognized atom names and non-integral net charges are reported in the [Reply Log](#); such problems should be examined and [resolved](#). Note that chain-terminal nucleotide residues will normally have non-integral charges, but the 5' and 3' charges sum to an integer.



CHIMERA

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Atom Specification

Atoms may be specified in [commands](#) using:

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or [combinations](#) of these. Some commands accept specifications of [non-atomic models](#).

In many commands where the specification is the last argument (e.g. [color](#)), a blank specification means "all."

In commands where the specification is not necessarily the last argument (e.g. [findclash](#)), specifications with embedded spaces should be enclosed in single or double quote marks.

When a plus sign (+) has been typed into the [Command Line](#), it will be replaced by the atom specification string of the next [picked](#) atom. Each plus sign must be preceded and followed by a space (or the end of the line). In addition, the following are valid atom specifications:

- **selected**, **sel**, or **picked**, indicating atoms that are [selected](#)
- names of previously [saved selections](#), using **sel=selection_name** or just *selection_name*; if a saved selection has the same name as a [surface category](#), the saved selection will be used.

Those familiar with atom specification in Midas may wish to consult the [summary of differences](#).

Hierarchical Specifiers

BASIC ATOM SPECIFICATION		
Symbol	Reference Level	Definition
#	model	number assigned to the model by default or by the user with the open command.
:	residue	residue name OR residue sequence number, with any insertion code appended
::	residue	residue name
@	atom	atom name

Each file of coordinates opened in Chimera becomes a *model* with an associated model ID number. Model numbers are assigned by the user with the [open](#) command or sequentially by default. Each model contains one or more *residues*, and each residue contains one or more *atoms* with names that are unique within that residue. Thus, an atom can be specified by its model number, residue number, and atom name. The lack of a specifier is interpreted to mean all units of the associated reference level; for example, if a model number is not given, the specification refers to all models.

Residue and atom names are read from the input file. In [PDB format](#), a standard nomenclature is used for standard amino acid and nucleic acid residues. Asterisks (*) in input atom names will be translated to prime symbols ('), but translated back if coordinates are later [saved](#) to a file. Prime symbols in input atom names are not translated on input or output.

Model and residue numbers are integers, although a residue number may have an insertion code directly appended. Multiple model numbers or residue numbers may be

indicated by comma-separated lists and/or one or more ranges of the form *start-end*. The words **start** and **end** can be substituted for *start* and *end*, respectively, and **all** can be substituted for the whole range (same as no specification at that level).

Residue names and numbers cannot be specified together using a single colon, but both can be specified in a single line by using more than one colon. For example, to specify all residues named HEM and residue 52:

:hem:52 (NOT **:hem,52**)

More examples:

#0

- all atoms in model 0

#3:45 - 83,90 - 98

- residues 45-83 and 90-98 in model 3

#0:12@CA,N

- alpha carbon and nitrogen of residue 12 in model 0

:12,14@CA

- alpha carbons of residues 12 and 14

:12:14@CA

- all atoms of residue 12 and alpha carbon of residue 14

:12 - 20@CA:14@N

- alpha carbons of residues 12-20 and nitrogen of residue 14

:lys

- or -

::lys

- all lysine residues

In some cases, the [basic notations](#) cannot uniquely specify the model, residue, or atom of interest:

SUBCATEGORIES	
General Form	Explanation
<i>#model(s).submodel(s)</i>	when a single PDB file contains multiple MODELS, they are considered submodels 1, 2, ... of a single model in Chimera
<i>:residue(s).chain(s)</i>	when a single model contains multiple chains, a unique specification includes both residue number and chain identifier
<i>@atom(s).altloc(s)</i>	when a single residue contains alternate locations of certain atoms, an independent specification includes both atom name and alternate location identifier

Submodel(s) are integers and may be indicated by a single value or a range of the form *start-end*. The words **start** and **end** can be substituted for *start* and *end*, respectively, and **all** can be substituted for the whole range (same as no submodel specification).

Chain(s) and *altloc(s)* are alphabetical characters. Any residue in PDB HETATM records (or the mmCIF equivalent) that does not already have a chain identifier is assigned to chain **het**, unless the residue is named WAT or HOH, in which case it is assigned to chain **water**; residue numbers are unchanged. (See also the [residue attribute isHet](#).)

Capitalization of residue and atom names, chain identifiers, insertion codes, or alternate location identifiers is not

important, with one exception: when a model contains both uppercase and lowercase chain identifiers, case matters for chain specification in that model only.

Subcategorizations are appended to the basic specification. The symbol for the relevant category (#, :, or @) must precede the subcategory specification, although they need not be immediately adjacent. **Because commas are used only to separate values of the basic reference levels (model, residue, and atom), they cannot be used to separate values of the sublevels directly.** For example,

#0.1 - 3,5

is interpreted as submodels 1-3 of model 0 and all of model 5, while

#0.1 - 3,.5

indicates submodels 1-3 of model 0 and submodel 5 of all models.

A subcategory specification applies only to the preceding category value(s) not separated from it by any commas. Thus,

:50.B,.D

indicates residue 50 in chain B and all residues in chain D;

:12 - 15,26 - 28.a,45.b@ca

specifies CA atoms in residues 12-15 in all chains (except the [special chains](#) **het** and **water**), 26-28 in chain A, and 45 in chain B. The following specifies residues 12-15 in no-ID chains only:

:12 - 15.

More examples:

:.b

- chain B

:a@n,ca,c,o

- peptide backbone atoms in chain A

:195.a,221.a@n,ca,c,o

- peptide backbone atoms in residues 195 and 221 of chain A

:522.water

- [water](#) residue 522 (a HETATM residue named HOH or WAT without a chain ID in the input file)

@.a

- all atoms with alternate location identifier A

WILD CARDS

The **global wild card *** matches all atoms in a residue or all residues in a model. It stands alone as a symbol, that is, it cannot be used to match *parts* of names, such as **G*** or ***A**. The **partial wild card =** matches parts of atom or residue names but not parts of residue sequence numbers; similarly, the **single-character wild card ?** matches single characters within residue or atom names but not single digits within residue sequence numbers. For example:

#1:12@*

- or -

#1:12

- all atoms in residue 12 of model 1

#0,1,2:50-*@CA

- all alpha carbon atoms in residues 50 to the end of models

0, 1 and 2

#2:G??

- all residues which have three-letter names beginning with **G** in model 2

:fmn@?1

- atoms within residue FMN which have two-letter names ending with **1**

@S=

- all atoms which have names beginning with **S**; in general, this will be all sulfur atoms

#0@H@H?@H??

- all atoms with one-, two-, or three-letter names beginning with **H** in model 0

Zones

Zone specifiers indicate atoms and residues that are within or beyond a given distance of the referenced atom(s). **z<** and **zr<** specify all *residues* with any atom within the given distance from the referenced atoms. **za<** specifies all *atoms* within the given distance. **z>**, **zr>**, and **za>** yield the sets complementary to their **<** counterparts. For example,

#1:gtp za<10.5

specifies all atoms within 10.5 Å of any atom in residue GTP, model 1.

Zone specifiers refer to atoms only, not surfaces; however, the command [zonesel](#) allows basing zones on surfaces and including surfaces in zones.

Built-in Classifications

Any of the entries in the following sections of the [Select menu](#) can be used for command-line atom specification:

- **Chemistry**
- **Residue...** [amino acid category](#)
- **Structure**

For example, **Structure** includes

- type of molecule: **protein**, **nucleic acid**, [markers](#)
- protein secondary structure: **helix**, **strand**, **coil**
- [automatic categories for surface calculation](#): **main**, **ligand**, **solvent**, **ions**

The same pattern of capitalization and spaces, if any, as shown in the [Select menu](#) should be used. Where there is ambiguity, the parent menu should be included in the specification, for example, "IDATM type.H" or "element.H" instead of "H" alone. The parent menu can be included in this manner even when not necessary.

One can also use custom amino acid categories that have been defined with [ResProp](#) and custom categories for surface calculation that have been defined with the command [msms cat \(surfcats\)](#). If a [saved selection](#) has the same name as a surface category, the saved selection will be used.

Some examples:

side chain/base.without CA/C1'

- or -

without CA/C1'

- atoms in amino acid side chains (not including CA) and atoms in nucleic acid bases (not including C1')

#1 & Mg

- magnesium atoms/ions in model 1

helix & positive

- residues in helices that are also in the **positive** [amino acid category](#)

carboxylate

- atoms in carboxylate groups

solvent

- atoms [automatically categorized](#) as **solvent**
-

Attributes

[Attributes](#) are properties of atoms, residues and models. The **slash mark /** indicates specification by [attribute name](#) and value. The symbol for the relevant category (@ for [atom attributes](#), : for [residue attributes](#), # for [model attributes](#)) **must precede the slash mark**, although it need not be immediately adjacent.

Multiple attributes at the same reference level (different atom properties, for example) can follow a single slash mark and should be separated by **and** or **or**. When **and** and **or** occur in the same list, **and** has higher priority (**and**-separated lists can be considered as grouped within parentheses).

The attribute *names* are case-sensitive; the attribute *values*, if any, are case-sensitive if specified with **==**, but not if specified with **=**. Attribute values containing spaces (some color names, for example) must be enclosed by double quotes. The **exclamation mark !** indicates that the atoms, residues, or models must **not** match the subsequent attribute

specification. For yes/no properties the syntax is !
attribute_name, and for multivalued properties the syntax is
attribute_name!=value.

Attributes with numerical values can also be used with > (greater than), < (less than), >= (greater than or equal to), and <= (less than or equal to).

Color values can be specified:

- by name, [built-in](#) or defined previously with [colordef](#)
- as a comma-separated list of component values *red, green, blue, alpha* where each value can range from zero to 1 and specification of *alpha* (1 – transparency) is optional

When placed before an attribute name, the **caret** ^ indicates that the atoms, residues, or models have not been assigned any value for the attribute. For example, *:/*

^kdHydrophobicity designates residues (such as water or nucleic acids) that lack a [Kyte-Doolittle hydrophobicity](#) assignment.

The operators ~ and !~ can be used instead of = and !=, respectively, to indicate that the subsequent string should be treated as a [regular expression](#).

SELECTED ATOM ATTRIBUTES ^{**}	
Name and Usage	Explanation
altLoc = <i>altloc</i>	<i>altloc</i> is the alternate location identifier of the atom
areaSAS = <i>sasa</i>	<i>sasa</i> is the solvent-accessible surface area of the atom (available when a molecular surface has been computed)

areaSES = <i>sesa</i>	<i>sesa</i> is the solvent-excluded surface area of the atom (available when a molecular surface has been computed)
bfactor = <i>bfactor</i>	<i>bfactor</i> is the B-factor value of the atom (see also Thermal Ellipsoids)
color = <i>color</i>	<i>color</i> is the color of the atom (assigned on a per-atom basis; see coloring hierarchy)
defaultRadius = <i>rad</i>	<i>rad</i> is the default VDW radius of the atom in Å
display	whether display is enabled at the atom level (see display hierarchy)
drawMode = <i>mode</i>	<i>mode</i> can be 0 (synonyms: dot, wire, wireframe), 1 (sphere, cpk, space-filling), 2 (endcap, stick), or 3 (ball, ball and stick, ball-and-stick, ball+stick, bs, b+s); see draw mode
element = <i>atno</i>	<i>atno</i> is the atomic number
idatmType = <i>type</i>	<i>type</i> is the atom type
label	whether the atom is labeled
label = <i>label</i>	<i>label</i> is the text of the atom label
labelColor = <i>labcolor</i>	<i>labcolor</i> is the color of the atom label
name = <i>name</i>	<i>name</i> is the atom name
occupancy = <i>occupancy</i>	<i>occupancy</i> is the occupancy value of the atom

radius= <i>radius</i>	<i>radius</i> is the current VDW radius of the atom in Å (may have been changed by the user from the default VDW radius)
serialNumber= <i>n</i>	<i>n</i> is the atom serial number in the input file
surfaceCategory= <i>category</i>	<i>category</i> is the name of the surface calculation category to which the atom has been assigned automatically or manually using surfcats
surfaceDisplay	whether molecular surface display is turned on for the atom (however, this can be true even for atoms that do not contribute to the molecular surface)

Examples:

@ca/!label and color!=green and color!=red

- or -

@/name=ca and !label and color!=green and color!=red

- atoms named CA which are not labeled, and are not green or red

@n/drawMode=1 and color=green

- atoms named N that are green and drawn as spheres

@n/drawMode=1 or color=green

- atoms named N that are green and/or drawn as spheres

@/color=yellow or color=blue and label

- atoms that are yellow and atoms that are both blue and

labeled

@/color!=yellow or color!=blue

- all atoms, because if an atom is yellow it fulfills the criterion of not being blue, and *vice versa*

@/bfactor>=50

- atoms with B-factor values greater than or equal to 50

@/bfactor>=20 and bfactor<=40

- atoms with B-factor values ranging from 20 to 40

SELECTED RESIDUE ATTRIBUTES ^{**}	
Name and Usage	Explanation
areaSAS= <i>sasa</i>	<i>sasa</i> is the solvent-accessible surface area of the residue (available when a molecular surface has been computed)
areaSES= <i>sesa</i>	<i>sesa</i> is the solvent-excluded surface area of the residue (available when a molecular surface has been computed)
isHelix	whether the residue is in a helix (true is only possible for amino acids)
isHet	whether the residue is in PDB HETATM records (or the mmCIF equivalent)
isSheet OR isStrand	whether the residue is in a beta strand (true is only possible for amino acids)
kdHydrophobicity= <i>value</i>	<i>value</i> is the Kyte-Doolittle hydrophobicity of the amino acid residue

numAtoms = <i>N</i>	<i>N</i> is the total number of atoms in the residue
phi = <i>angle</i>	<i>angle</i> is the protein/peptide backbone ϕ dihedral angle (C_{i-1} -N-CA-C)
psi = <i>angle</i>	<i>angle</i> is the protein/peptide backbone ψ dihedral angle (N-CA-C- N_{i+1})
ribbonColor = <i>ribcolor</i>	<i>ribcolor</i> is the color of the residue's ribbon segment (see coloring hierarchy)
ribbonDisplay	whether ribbon display is turned on for the residue (however, this can be true even when the residue is a type that does not have any ribbon, such as water)
ssId = <i>N</i>	<i>N</i> is the secondary structure element identifier, for example, 1 for residues in the first helix and first strand (starting from the N-terminus)
type = <i>resname</i>	<i>resname</i> is the residue name
uniprotIndex = <i>N</i>	<i>N</i> is the residue number in the corresponding UniProt sequence (discerned from HEADER and SEQRES information in the structure PDB file using a web service provided by the RCSB PDB)

Helix and strand assignments are taken from the input structure file. When the input file lacks secondary structure information, [ksdssp](#) is called automatically to generate helix and strand assignments. The [ksdssp](#) command (or [compute SS](#) in the [Model Panel](#)) can also be used to recompute assignments.

Examples:

:/type!=gly and type!=pro

- all residues not named GLY or PRO

:/isStrand :/isHelix

- or -

:/isStrand or isHelix

- all amino acid residues in beta strands or helices

:/isStrand and isHelix

- nothing, because the criteria are mutually exclusive

SELECTED MOLECULE MODEL ATTRIBUTES ^{**}	
Name and Usage	Explanation
ballScale = <i>factor</i>	<i>factor</i> is ball radius relative to VDW radius
color = <i>color</i>	<i>color</i> is the color assigned on a per-model basis (see coloring hierarchy)
display	whether display is enabled at the model level (see display hierarchy)
lineWidth = <i>width</i>	<i>width</i> is the linewidth of bonds in the model in the wire representation
numAtoms = <i>N</i>	<i>N</i> is the total number of atoms in the model
numResidues = <i>M</i>	<i>M</i> is the total number of residues in the model
ribbonInsideColor = <i>color</i>	<i>color</i> is the color used for the insides of peptide/protein helix ribbon segments

stickScale = <i>factor</i>	<i>factor</i> is stick radius relative to bond radius (the default bond radius is 0.2 Å)
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** Additional [attributes](#) can be created:

- arbitrarily with [Define Attribute](#), [defattr](#), or [setattr](#)
- by combining other attributes using the [Attribute Calculator](#)
- by various other Chimera tools such as [Add Charge](#), [Area/Volume from Web](#), [Values at Atom Positions](#), and [Multalign Viewer](#)

Attribute Names

The preceding tables [[atoms](#)] [[residues](#)] [[molecule models](#)] include the names of some commonly used [attributes](#). More obscure attributes are omitted, but more importantly, the tables are incomplete because some attributes may be generated only when a particular tool is used, or created arbitrarily by the user.

Name lookup in Chimera:

- attribute names are shown in the balloon help of [attribute inspector](#) dialogs
- [Select by Attribute](#) lists most numerical, string-valued, and boolean attributes of atoms, residues, and molecule models; [Render by Attribute](#) lists most numerical ones
- the command [list resattr](#) lists most residue attributes
- Data descriptors (attributes) defined in the C++ layer of Chimera can be listed by entering the following into [IDLE](#) (under **Tools... General Controls**):

help(chimera.*Class*)

where *Class* can be **Atom**, **Residue**, or **Molecule**. This approach also works for **Bond**, **PseudoBond**, and **PseudoBondGroup**. Attributes defined in the Python layer will not be included, however.

Combinations

Atom specifications can be combined with the operators:

- **&** for **intersection** (AND)
- **|** for **union** (OR)
- **~** for **negation** (NOT)

Space-delimiting these operators is optional. When **&** and **|** occur in the same list, **&** has higher priority (**&**-separated lists can be considered as grouped within parentheses).

Note that a different set of operators (**and**, **or**, **!**, *etc.*) are used to combine [attribute](#) tests; however, both types of operators can be used within the same specification.

Examples:

#1:/type=asp or type=glu & #0 z<10

- or -

#1:asp,glu & #0 z<10

- aspartate residues and glutamate residues in model 1 that are within 10 Å of model 0

:cys@sg & ~disulfide

- or -

:cys&S&~disulfide

- cysteine sulfur atoms not participating in a disulfide bond

ions za<4 & ~ ions

- atoms within 4 Å of atoms [categorized](#) as ions, excluding the ions themselves

ligand z<5 &~ ligand &~ solvent

- residues with any atoms within 5 Å of residues [categorized](#) as ligand, excluding ligand and solvent

Ng+|N3+

- guanidinium nitrogens and sp^3 -hybridized, formally positive nitrogens (see [atom types](#))

Surfaces and Other Model Types

Some commands can apply to things other than molecule models (atomic coordinates) and [molecular surfaces](#). To handle such cases, "atom specification" has been extended as follows:

- an entire [surface model](#) (all of its pieces) can be specified by model number preceded by #, or by [selection](#) and using the word **selected**, **sel**, or **picked**
- [surface pieces](#) can be specified individually by [selection from the screen](#) and using the word **selected**, **sel**, or **picked**
- [other types of models](#) can be specified by model number preceded by #

Which of these can be used and whether they can be [combined](#) with atom specifications depends on the command.

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